

FC7xxx Mailbox User Manual

Rev.0.6

Revision History

Revision	Date	Changes
0.1	2023/10/20	Initial release for MCAL V0.3.0
0.6	2023/03/29	Updated for MCAL V0.6.0 - Added support for FC7240

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Chapter 1 Mailbox Introduction

1.1 Requirements

Mailbox (Mb) is a Complex Device Driver (CDD), so there are no AUTOSAR requirements regarding this module. It has vendor-specific requirements and implementation. Mailbox driver has two usage modes.

- **Free Mode**
This mode configures a mailbox channel where the request and done events are specified to be received by one or multiple cores. Each core allowed to use the channel can send two words of custom data or trigger a done event through the channel. The transfer process is zero-copy.
- **Message Queue**
This mode implements a message queue between multiple cores using a mailbox channel. Each message queue can only specify one sender and one receiver. The sender writes the message to the message pool and notifies the receiver, while the receiver can get data from the message pool.

1.2 Hardware Summary

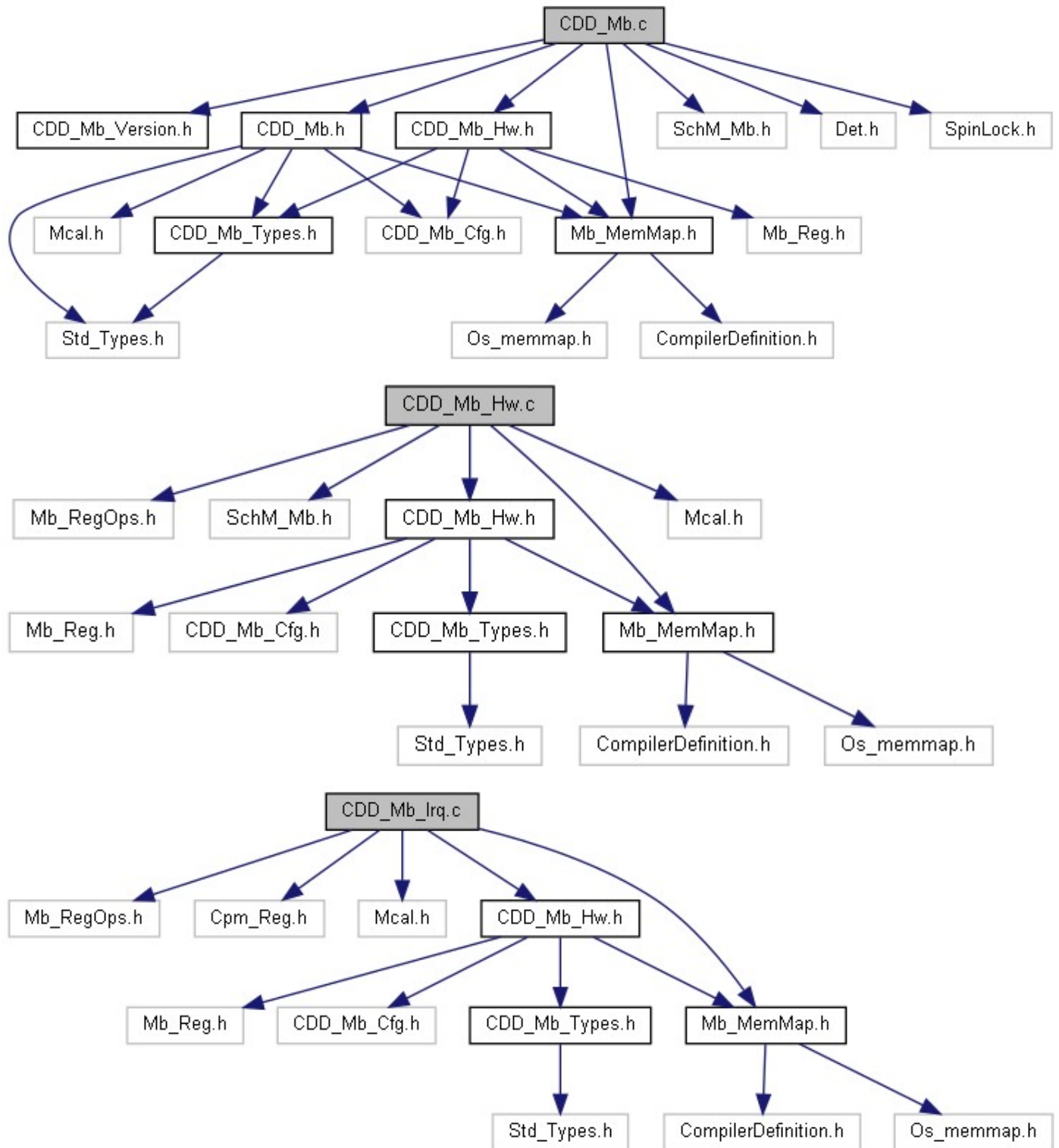
The Mailbox driver is implemented as an Autosar complex device driver. provides the functionality for multiple cores to communicate and synchronize their activities

Chapter 2 Software Design

2.1 Rejected Requirements

None

2.2 File Structure



2.3 Macros

2.3.1 Macros in CDD_Mb.h

- **#define MB_E_UNINIT** ((uint8)0x0AU)
API service used without Mailbox module initialization.
- **#define MB_E_ALREADY_INITIALIZED** ((uint8)0x0BU)
The Mailbox module is already initialized
- **#define MB_E_PARAM_CONFIG** ((uint8)0x0CU)
The Mailbox module is not properly configured.
- **#define MB_E_PARAM_CHANNEL** ((uint8)0x0DU)
API service used with an invalid channel identifier or channel was not configured for the functionality of the calling API.
- **#define MB_E_PARAM_CORE** ((uint8)0x0E)
API service used with an invalid core identifier or core was not configured for the functionality of the calling API.
- **#define MB_E_CHANNEL_LOCKED** ((uint8)0x10)
API service used with an invalid channel identifier or channel has already been locked.
- **#define MB_E_CHANNEL_UNLOCKED** ((uint8)0x11)
API service used with an invalid channel identifier or channel is not locked.
- **#define MB_E_MESSAGE_QUEUE_FULL** ((uint8)0x13)
Message queue is full.
- **#define MB_E_MESSAGE_QUEUE_EMPTY** ((uint8)0x14)
Message queue is empty.
- **#define MB_E_MESSAGE_QUEUE_BUSY** ((uint8)0x15)
Message queue is busy.
- **#define MB_E_GET_SPIN_LOCK_FAILED** ((uint8)0x16)
GET Spin Lock failed.
- **#define MB_E_PARAM_VINFO** ((uint8)0x19)
API Mb_GetVersionInfo is called and the parameter versioninfo is invalid (e.g. NULL).
- **#define MB_E_INV_PARTITION** ((uint8)0x1A)
API service partition mapping check error.

- **#define MB_INIT_ID 0x00U**
API service ID for Mb_Init function.
- **#define MB_DEINIT_ID 0x01U**
API service ID for Mb_DeInit function.
- **#define MB_SEND_DATA_ID 0x02U**
API service ID for Mb_SendData function.
- **#define MB_DONE_CHANNEL_ID 0x03U**
API service ID for Mb_DoneChannel function.
- **#define MB_SEND_MESSAGE_ID 0x04U**
API service ID for Mb_SendMessage function.
- **#define MB_GET_MESSAGE_ID 0x05U**
API service ID for Mb_GetMessage function.
- **#define MB_GET_MESSAGE_COUNT_ID 0x06U**
API service ID for Mb_GetMessageCount function.
- **#define MB_GET_MESSAGE_QUEUE_STATE_ID 0x07U)**
API service ID for Mb_GetMessageQueueState function.
- **#define MB_RESET_MESSAGE_QUEUE_ID 0x08U**
API service ID for Mb_ResetMessageQueue function.
- **#define MB_RESET_CHANNEL_ID 0x09U**
API service ID for Mb_ResetChannel function.
- **#define MB_GET_CHANNEL_STATE_ID 0x0AU**
API service ID for Mb_GetChannelStatus function.
- **#define MB_GETVERSIONINFO_ID 0x0BU**
API service ID for Mb_GetVersionInfo function.

2.3.2 Macros in CDD_Mb_Version.h

- **#define MB_AR_RELEASE_MAJOR_VERSION 4**
- **#define MB_AR_RELEASE_MINOR_VERSION 6**
- **#define MB_AR_RELEASE_REVISION_VERSION 0**

- #define **MB_SW_MAJOR_VERSION** 0
- #define **MB_SW_MINOR_VERSION** 6
- #define **MB_SW_PATCH_VERSION** 0
- #define **MB_VENDOR_ID** 174
- #define **MB_MODULE_ID** 254

2.3.3 Macros in CDD_Mb_Cfg.h

- #define **MB_CFG_AR_RELEASE_MAJOR_VERSION** 4
- #define **MB_CFG_AR_RELEASE_MINOR_VERSION** 6
- #define **MB_CFG_AR_RELEASE_REVISION_VERSION** 0
- #define **MB_CFG_SW_MAJOR_VERSION** 0
- #define **MB_CFG_SW_MINOR_VERSION** 6
- #define **MB_CFG_SW_PATCH_VERSION** 0
- #define **MB_CFG_VENDOR_ID** 174
- #define **MB_DEV_ERROR_DETECT** (STD_ON/STD_OFF)
Mb Dev error detect switch.
- #define **MB_DEINIT_API** (STD_ON/STD_OFF)
MbDeInitApi switch.
- #define **MB_MESSAGE_QUEUE_API** (STD_ON/STD_OFF)
MbMessageQueueApi switch.
- #define **MB_RESET_CHANNEL_API** (STD_ON/STD_OFF)
MbResetChannelApi switch.
- #define **MB_GET_CHANNEL_STATUS_API** (STD_ON/STD_OFF)
MbGetChannelStatusApi switch.
- #define **MB_GET_VERSION_INFO_API** (STD_ON/STD_OFF)
MbGetVersionInfoApi switch.

- #define **MB_MAX_HW_CHANNEL** (XU)
Maximum number of Mailbox hardware channel.
- #define **MB_MAX_HW_INTR_CHANNEL** (XU)
Maximum number of Mailbox hardware interrupt channel.
- #define **MB_MAX_CHANNEL** (XU)
Maximum number of Mailbox channels configured.
- #define **MB_FORCE_UNLOCK_CODE** (XU)
Master ID for each core.
- #define **MB_MASTER_ID_TABLE** {X,X}
Command to force unlock a mailbox channel.
- #define **MB_SPIN_LOCK_TIMEOUT** (XU)
The timeout for acquiring a spin lock
- #define **MB_FREE_CHANNEL_COUNT** (XU)
Number of Mailbox channels configured as free channel.
- #define **MB_MESSAGE_QUEUE_COUNT** (XU)
Number of Mailbox channels configured as message queue.
- #define **MB_CORE_COUNT** (XU)
The max number of core used by mailbox.

2.3.4 Macros in CDD_Mb_Hw.h

- #define **MB_HW_UNUSED** (0xFFU)
Mailbox hardware channel unused.

2.4 Enums

2.4.1 Enumerations in CDD_Mb.h

2.4.1.1 Mb_ReturnStateType

Enumeration	Mb_ReturnStateType	
Description	Allow enabling / disabling of all interrupts which are not required for the ECU wakeup.	
Values	Value	Description
	MB_RET_STATE_SUCCESS	The operation is successful.
	MB_RET_STATE_FAILED	The operation is failed.
	MB_RET_STATE_BUSY	The channel is busy.

	MB_RET_STATE_FULL	The message queue is full.
	MB_RET_STATE_EMPTY	The message queue is empty.

2.4.2 Enumerations in CDD_Mb_Types.h

2.4.2.1 Mb_CoreType

Enumeration	Mb_CoreType	
Description	Numeric identifier of Cores.	
Values	Value	Description
	MB_CORE_0	The destination/source of the data is CORE0.
	MB_CORE_1	The destination/source of the data is CORE1.
	MB_CORE_2	The destination/source of the data is CORE2.
	MB_CORE_HSM	The destination/source of the data is HSM.

2.4.2.2 Mb_FreeChannelConfigFlagType

Enumeration	Mb_FreeChannelConfigFlagType	
Description	Configuration parameters for a free channel, each bit represents a corresponding configuration	
Values	Value	Description
	MB_FREE_CH_CFG_REQ_TO_CORE0	Core0 can receive request events issued by this channel.
	MB_FREE_CH_CFG_REQ_TO_CORE1	Core1 can receive request events issued by this channel.
	MB_FREE_CH_CFG_REQ_TO_CORE2	Core2 can receive request events issued by this channel.
	MB_FREE_CH_CFG_DONE_TO_CORE0	Core0 can receive done events issued by this channel.
	MB_FREE_CH_CFG_DONE_TO_CORE1	Core1 can receive done events issued by this channel.
	MB_FREE_CH_CFG_DONE_TO_CORE2	Core2 can receive done events issued by this channel.
	MB_CHANNEL_AUTO_DONE	The receiver automatically send done event after receiving the message.

2.4.2.3 Mb_ChannelModeType

Enumeration	Mb_ChannelModeType	
Description	Definition of the channel mode.	
Values	Value	Description
	MB_MODE_FREE	The current channel is idle.
	MB_MODE_MESSAGE_QUEUE	This channel operates in Message Queue mode.

2.4.2.4 Mb_MessageQueueStateType

Enumeration	Mb_MessageQueueStateType	
Description	Definition of the message queue state.	
Values	Value	Description
	MB_MESSAGE_QUEUE_EMPTY	The message queue is empty.
	MB_MESSAGE_QUEUE_FULL	The message queue is full.
	MB_MESSAGE_QUEUE_HALF_FULL	The message queue is half full.

2.4.2.5 Mb_ChannelStateType

Enumeration	Mb_ChannelStateType	
Description	Definition of mailbox channel state.	
Values	Value	Description
	MB_CHANNEL_IDLE	The current channel is idle.
	MB_CHANNEL_BUSY	The current channel is busy.

2.5 Typedefs

2.5.1 Typedefs in CDD_Mb_Types.h

- typedef uint32 Mb_ChannelType
Numeric identifier of an Mailbox channel.

2.6 Structures

2.6.1 Mb_RingBlockType

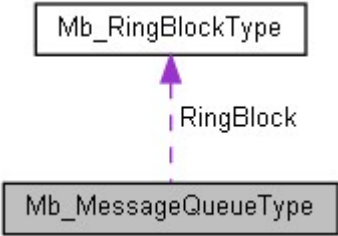
Structure	Mb_RingBlockType
Description	Configuration parameters for a ring block.
Diagram	N/A
Data Fields	- const uint32 u32BlockNum The number of ring blocks. - const uint32 u32BlockSize The size of each block. - uint8 * pBuffer The pointer to the memory pool of the ring block.

2.6.2 Mb_FreeChannelConfigType

Structure	Mb_FreeChannelConfigType
Description	The special configuration for a channel configured in free channel mode.
Diagram	N/A
Data Fields	- uint32 u32ConfigFlag Channel configuration parameters. Relevant Mb_FreeChannelConfigFlagType. - void(* pRequestCallback)(Mb_ChannelType Channel, Mb_CoreType MasterCore, uint32 u32Data0, uint32 u32Data1) Callback for request. - void(* pDone)(Mb_ChannelType Channel) Callback for done.

2.6.3 Mb_MessageQueueType

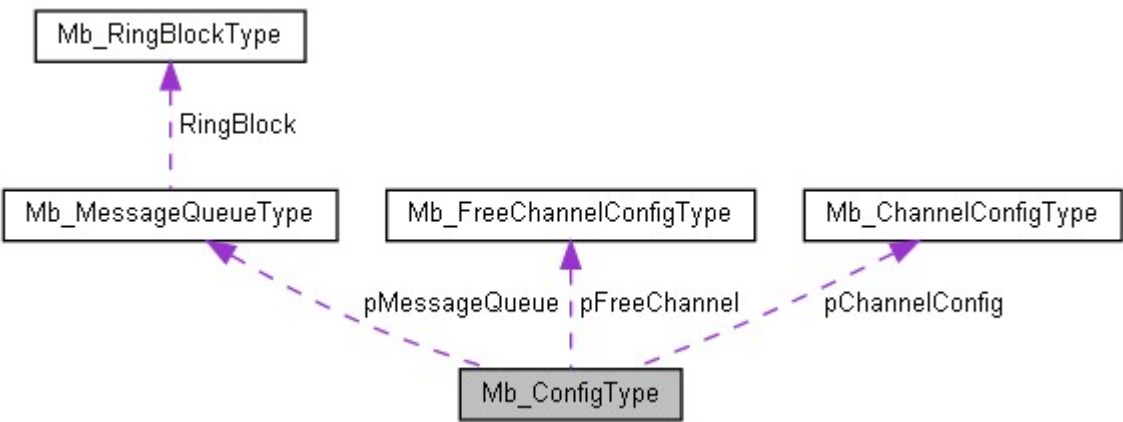
Structure	Mb_MessageQueueType
Description	The special configuration for a channel configured in message queue mode.

Diagram	 <pre> graph BT Mb_MessageQueueType -- RingBlock --> Mb_RingBlockType </pre>
Data Fields	• Mb_CoreType SenderCore The sender of messages. • Mb_CoreType ReceiverCore The receiver of messages. • Mb_RingBlockType RingBlock The ring block of the message queue. • void(* pReceivedCallback)(void) Callback for receive.

2.6.4 Mb_ChannelConfigType

Structure	Mb_ChannelConfigType
Description	The common configuration of a mailbox channel.
Diagram	N/A
Data Fields	• Mb_ChannelModeType eChannelMode The mode of the channel. • uint16 u16MbHwChannel The hardware channel used by the channel. • uint16 u16ChannelPartition The partitions that can use the channel.

2.6.5 Mb_ConfigType

Structure	Mb_ConfigType
Description	Structure for Configuration data.
Diagram	 <pre> graph BT Mb_ConfigType -- pMessageQueue --> Mb_MessageQueueType Mb_ConfigType -- pFreeChannel --> Mb_FreeChannelConfigType Mb_ConfigType -- pChannelConfig --> Mb_ChannelConfigType Mb_MessageQueueType -- RingBlock --> Mb_RingBlockType </pre>
Data Fields	• Mb_ChannelType tChannelCount The maximum number of configured channels. • Mb_ChannelConfigType * pChannelConfig The general configuration of channels.

	- Mb_FreeChannelConfigType * pFreeChannel The general configuration of channel The specific configuration of channels configured as a free channel. - Mb_MessageQueueType * pMessageQueue The specific configuration of channels configured as a message queue - Mb_ChannelType * pMbSubChannelMap Map of channel IDs to the index of the respective type. - Mb_ChannelType * pMbHwChannelMap Map of hardware index to channel ID.
--	---

2.7 API Functions

2.7.1 Functions in CDD_Mb.h

2.7.1.1 void Mb_Init (const Mb_ConfigType * ConfigPtr)

Function	void Mb_Init (const Mb_ConfigType * ConfigPtr)	
Description	Initializes the Mailbox channel and the driver.	
Diagram	 <pre> graph LR Mb_Init[Mb_Init] --> Mb_HL_Init[Mb_HL_Init] </pre>	
Parameters	Parameter	Description
	ConfigPtr	Pointer to configuration set in Variant PB (Variant PC requires a NULL_PTR).
Returns	N/A	

2.7.1.2 void Mb_DeInit (void)

Function	void Mb_DeInit (void)	
Description	De-initialize the Mailbox module.	
Diagram	 <pre> graph LR Mb_DeInit[Mb_DeInit] --> Mb_HL_DeInit[Mb_HL_DeInit] </pre>	
Parameters	N/A	
Returns	N/A	

2.7.1.3 Mb_ReturnStateType Mb_SendData (Mb_ChannelType Channel, Mb_CoreType DstCore, uint32 Data0, uint32 Data1)

Function	Mb_ReturnStateType Mb_SendData (Mb_ChannelType Channel, Mb_CoreType DstCore, uint32 Data0, uint32 Data1)	
Description	Sending data to a specific core through a mailbox channel.	
Diagram	 <pre> graph LR Mb_SendData[Mb_SendData] --> Mb_HL_SendData[Mb_HL_SendData] Mb_SendData --> Mb_ValidateChannelMode[Mb_ValidateChannelMode] </pre>	
Parameters	Parameter	Description
	Channel	Sending channel
	DstCore	Target core

	Data0	The DATA0 to be sent.
	Data1	The DATA1 to be sent.
Returns	MB_RET_STATE_SUCCESS The operation is successful. MB_RET_STATE_FAILED The operation is failed. MB_RET_STATE_BUSY The channel is busy with the current operation.	

2.7.1.4 void Mb_DoneChannel (Mb_ChannelType Channel)

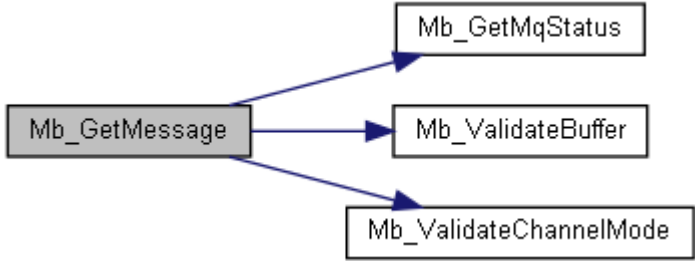
Function	void Mb_DoneChannel (Mb_ChannelType Channel)	
Description	Send a "done" event to the specified channel.	
Diagram	<pre> graph LR Mb_DoneChannel --> Mb_HL_DoneChannel Mb_DoneChannel --> Mb_ValidateChannelMode Mb_HL_DoneChannel --> Mb_HL_GetCoreIndex </pre>	
Parameters	Parameter	Description
	Channel	Channel to be done
Returns	N/A	

2.7.1.5 Mb_ReturnStateType Mb_SendMessage (Mb_ChannelType Channel, uint8 *Buffer, uint32 BufferLength)

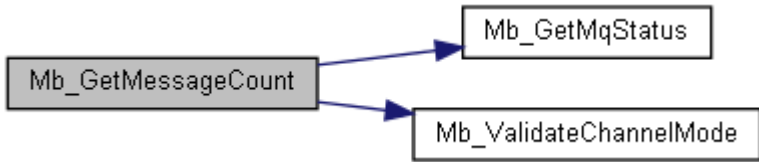
Function	Mb_ReturnStateType Mb_SendMessage (Mb_ChannelType Channel, uint8 *Buffer, uint32 BufferLength)	
Description	Send the message via the mailbox channel.	
Diagram	<pre> graph LR Mb_SendMessage --> Mb_GetMqStatus Mb_SendMessage --> Mb_HL_SendData Mb_SendMessage --> Mb_ValidateBuffer Mb_SendMessage --> Mb_ValidateChannelMode </pre>	
Parameters	Parameter	Description
	Channel	Channel to send the message
	Buffer	Pointer to the send buffer
	BufferLength	Send buffer byte length
Returns	MB_RET_STATE_FAILED The operation is failed. MB_RET_STATE_FULL The message queue is full. MB_RET_STATE_BUSY The channel is busy with the current operation.	

2.7.1.6 void Mb_GetMessage (Mb_ChannelType Channel, uint8 *ReceiveBuffer, uint32 ReceiveBufferLength)

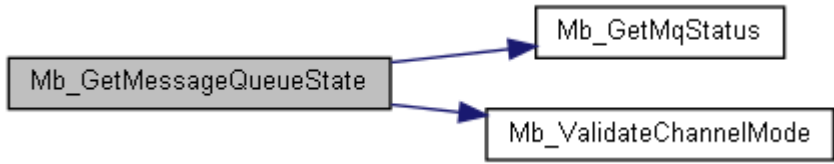
Function	void Mb_GetMessage (Mb_ChannelType Channel, uint8 *ReceiveBuffer, uint32 ReceiveBufferLength)	
Description	Get a message in message queue.	

Diagram		
Parameters	Parameter	Description
	Channel	Channel to receive a message
	ReceiveBuffer	Pointer to the receive buffer
	ReceiveBufferLength	Receive buffer byte length
Returns	N/A	

2.7.1.7 uint32 Mb_GetMessageCount (Mb_ChannelType Channel)

Function	uint32 Mb_GetMessageCount (Mb_ChannelType Channel)	
Description	Get the number of messages in the message queue.	
Diagram		
Parameters	Parameter	Description
	Channel	The channel to be get.
Returns	The number of messages in the message queue.	

2.7.1.8 Mb_MessageQueueStateType Mb_GetMessageQueueState (Mb_ChannelType Channel)

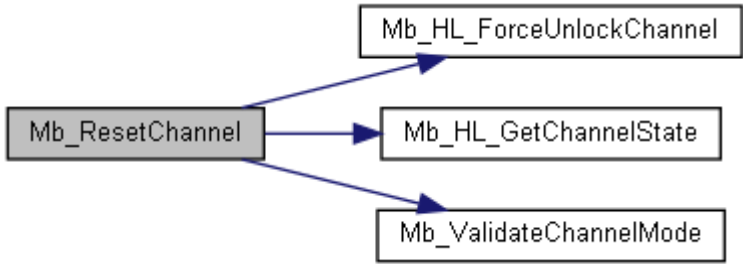
Function	Mb_MessageQueueStateType Mb_GetMessageQueueState (Mb_ChannelType Channel)	
Description	Get the current state of the message queue.	
Diagram		
Parameters	Parameter	Description
	Channel	The channel to be get.
Returns	MB_MESSAGE_QUEUE_EMPTY	The message queue is empty.
	MB_MESSAGE_QUEUE_FULL	The message queue is full.
	MB_MESSAGE_QUEUE_HALF_FULL	The message queue is half full.

2.7.1.9 Mb_ReturnStateType Mb_ResetMessageQueue (Mb_ChannelType Channel)

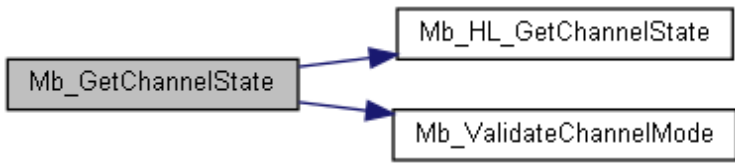
Function	Mb_ReturnStateType Mb_ResetMessageQueue (Mb_ChannelType Channel)	
Description	Clear all messages in the message queue.	
Diagram		
Parameters	Parameter	Description

	Channel	The channel to be cleared.
Returns	MB_RET_STATE_SUCCESS The operation is successful. MB_RET_STATE_BUSY The message queue is busy. MB_RET_STATE_FAILED The operation is failed.	

2.7.1.10 void Mb_ResetChannel (Mb_ChannelType Channel)

Function	void Mb_ResetChannel (Mb_ChannelType Channel)	
Description	Force reset the channel.	
Diagram	 <pre> graph LR Mb_ResetChannel[Mb_ResetChannel] --> Mb_HL_ForceUnlockChannel[Mb_HL_ForceUnlockChannel] Mb_ResetChannel --> Mb_HL_GetChannelState[Mb_HL_GetChannelState] Mb_ResetChannel --> Mb_ValidateChannelMode[Mb_ValidateChannelMode] </pre>	
Parameters	Parameter	Description
	Channel	Channel to reset
Returns	N/A	

2.7.1.11 Mb_ChannelStateType Mb_GetChannelState (Mb_ChannelType Channel)

Function	Mb_ChannelStateType Mb_GetChannelState (Mb_ChannelType Channel)	
Description	Get the channel state.	
Diagram	 <pre> graph LR Mb_GetChannelState[Mb_GetChannelState] --> Mb_HL_GetChannelState[Mb_HL_GetChannelState] Mb_GetChannelState --> Mb_ValidateChannelMode[Mb_ValidateChannelMode] </pre>	
Parameters	Parameter	Description
	Channel	Channel to get
Returns	MB_CHANNEL_BUSY The channel is busy. MB_CHANNEL_IDLE The channel is idle.	

2.7.1.12 void Mb_GetVersionInfo (Std_VersionInfoType *versioninfo)

Function	void Mb_GetVersionInfo (Std_VersionInfoType *versioninfo)	
Description	Returns the version information of this module.	
Diagram	N/A	
Parameters	Parameter	Description
	versioninfo	Pointer to where to store the version information of this module.
Returns	N/A	

2.8 Hardware Functions

2.8.1 Functions in CDD_Mb_Hw.h.

2.8.1.1 void Mb_HL_Init (uint16 u16ChannelRequestFlag, uint16 u16ChannelDoneFlag)

Function	void Mb_HL_Init (uint16 u16ChannelRequestFlag, uint16 u16ChannelDoneFlag)	
Description	This function initializes the Mailbox interrupt register.	
Diagram	N/A	
Parameters	Parameter	Description
	u16ChannelRequestFlag	Pointer to module configuration structure.
Returns	N/A	
Referenced By	Mb_Init()	

2.8.1.2 void Mb_HL_DeInit (void)

Function	void Mb_HL_DeInit(void)	
Description	This function de-initializes the Mailbox interrupt register.	
Diagram	N/A	
Parameters	N/A	
Returns	N/A	
Referenced By	void Mb_DeInit(void)	

2.8.1.3 Std_ReturnType Mb_HL_SendData(uint32 u32HwChannel, uint32 u32DstCore, uint32 u32Data0, uint32 u32Data1)

Function	Std_ReturnType Mb_HL_SendData (uint32 u32HwChannel, uint32 u32DstCore, uint32 u32Data0, uint32 u32Data1)	
Description	Send data via the mailbox channel.	
Diagram	N/A	
Parameters	Parameter	Description
	u32HwChannel	The hardware channel index
	u32DstCore	The index of the target core
	u32Data0	The data written to DATA0
	u32Data1	The data written to DATA1
Returns	E_OK	Write successful
	E_NOT_OK	Write failed, channel is busy
Referenced By	Mb_SendData(), Mb_SendMessage()	

2.8.1.4 Std_ReturnType Mb_HL_DoneChannel (uint32 u32HwChannel)

Function	Std_ReturnType Mb_HL_DoneChannel (uint32 u32HwChannel)	
Description	Trigger done event via the mailbox channel.	
Diagram	 <pre> graph LR A[Mb_HL_DoneChannel] --> B[Mb_HL_GetCoreIndex] </pre>	
Parameters	Parameter	Description
	u32HwChannel	Trigger done event via the mailbox channel.

Returns	E_OK Done successful
	E_NOT_OK Done failed, channel is idle or current core do not allow to trigger a done
Referenced By	Mb_DoneChannel(), Mb_RequestHandle()

2.8.1.5 Mb_ChannelStateType Mb_HL_GetChannelState(uint32 u32HwChannel)

Function	Mb_ChannelStateType Mb_HL_GetChannelState(uint32 u32HwChannel)	
Description	Get the current state of the channel.	
Diagram	N/A	
Parameters	Parameter	Description
	u32HwChannel	The hardware channel index
Returns	MB_CHANNEL_IDLE	The current channel is idle.
	MB_CHANNEL_BUSY	The current channel is busy.
Referenced By	Mb_GetChannelState(), Mb_ResetChannel()	

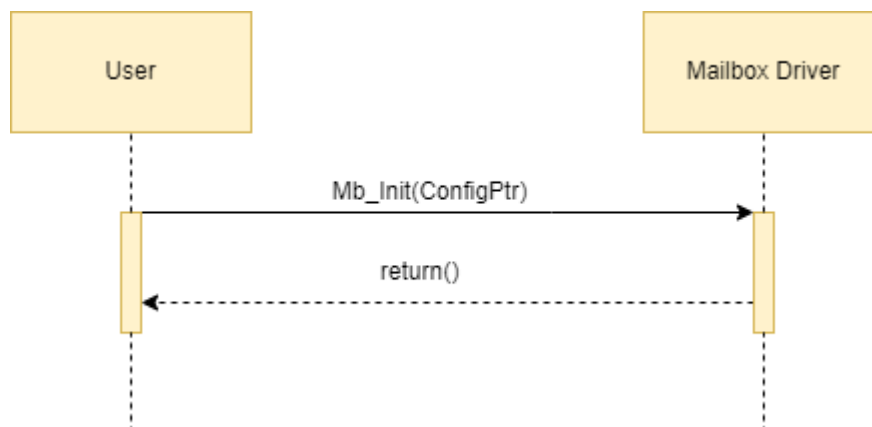
2.8.1.6 void Mb_HL_ForceUnlockChannel(uint32 u32HwChannel)

Function	void Mb_HL_ForceUnlockChannel (uint32 u32HwChannel)	
Description	Force unlock the channel.	
Diagram	N/A	
Parameters	Parameter	Description
	u32HwChannel	The hardware channel index
Returns	N/A	
Referenced By	Mb_ResetChannel()	

2.9 API Sequence Diagram

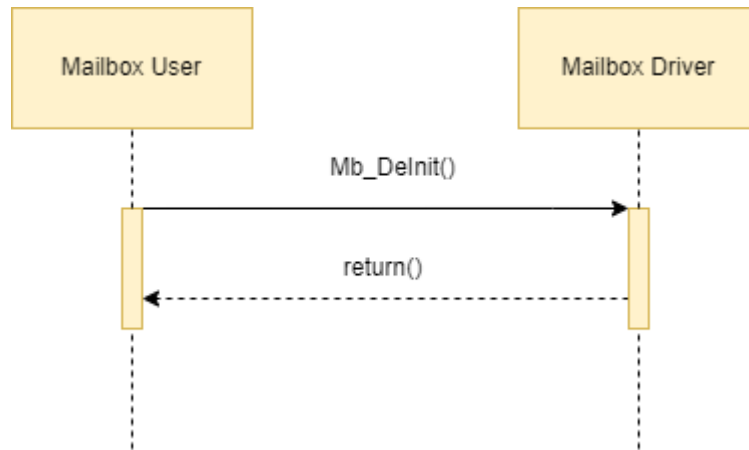
2.9.1 Initialization

Mailbox Driver Initialization



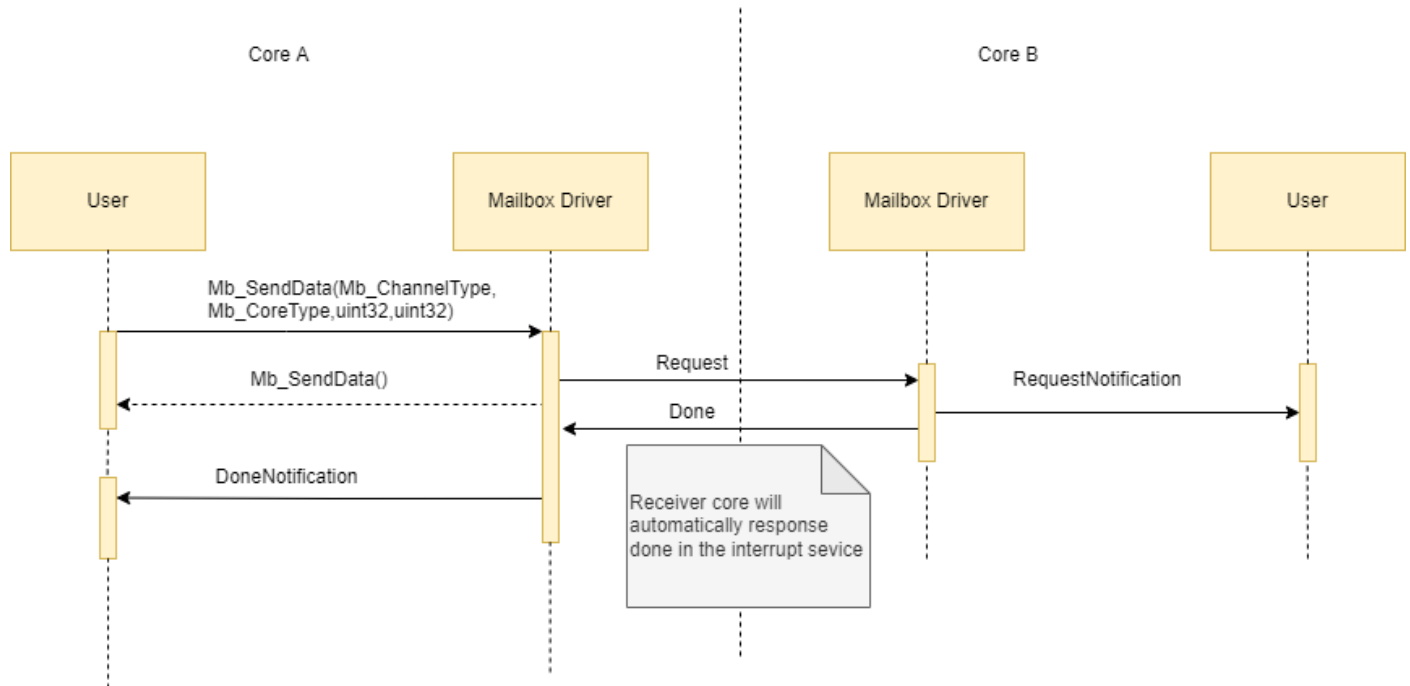
2.9.2 De-initialization

Mailbox Driver De-Initialization

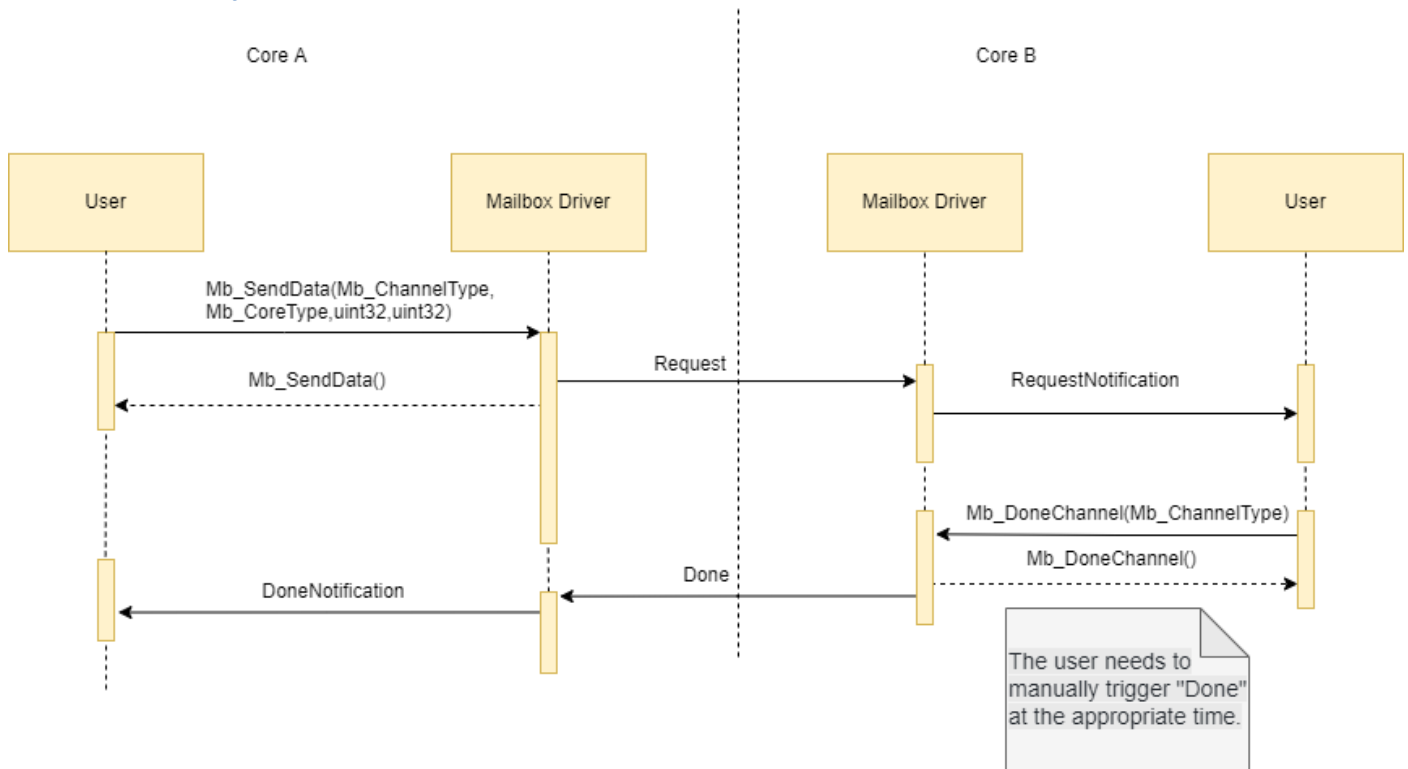


2.9.3 Free Channel Mode

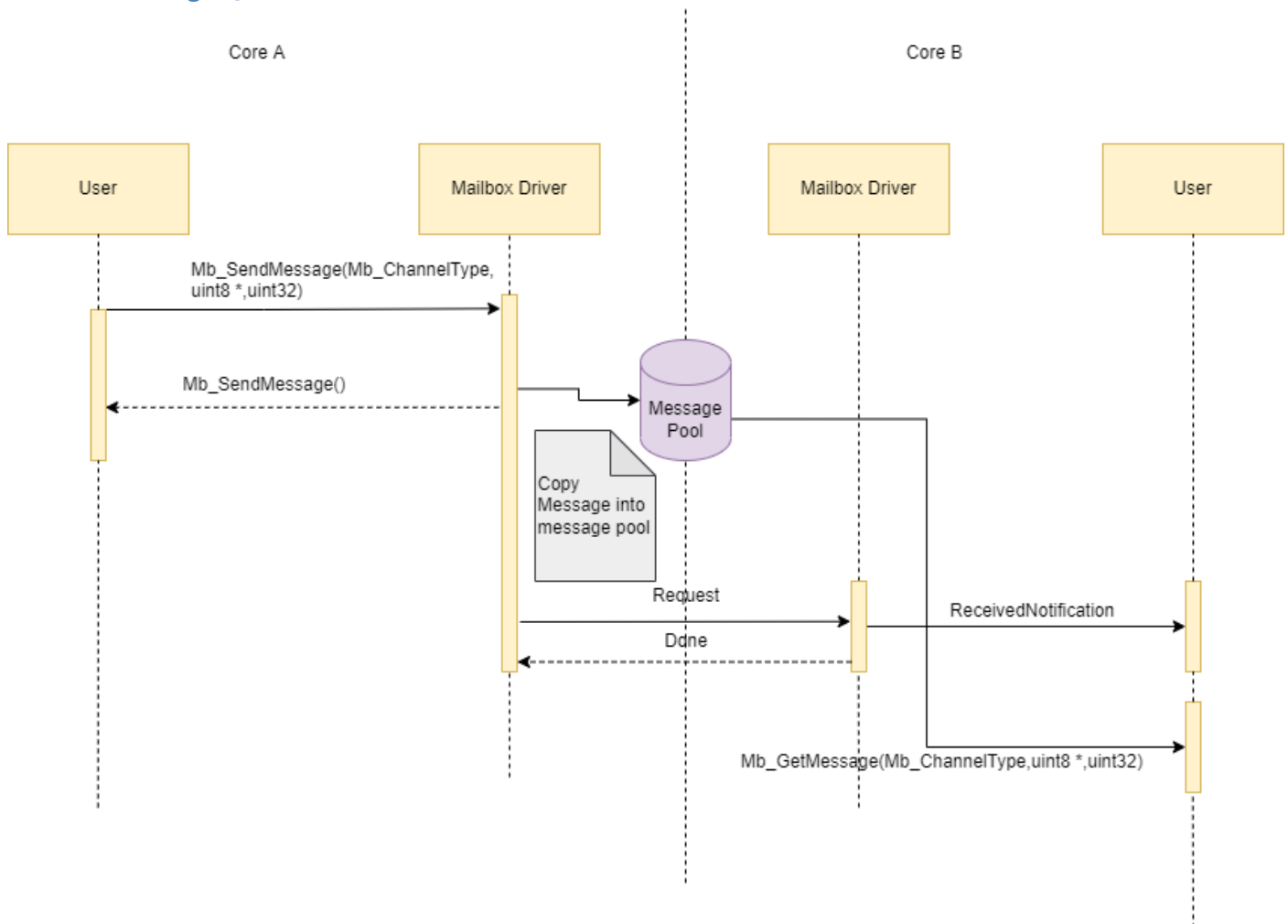
2.9.3.1 Automatically Done



2.9.3.2 Manually Done



2.9.4 Message Queue Mode



Chapter 3 Tresos Configuration Items

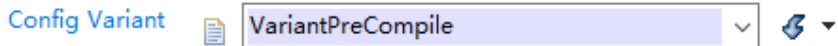
3.1 Container Inclusion Relation

The container inclusion relation is shown as below:



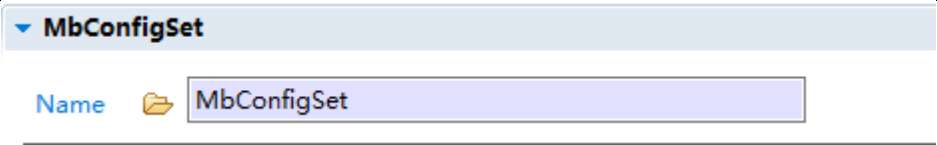
3.2 Containers and Variables

3.2.1 IMPLEMENTATION_CONFIG_VARIANT

Variable	IMPLEMENTATION_CONFIG_VARIANT		
Description	N/A		
Screenshot			
Properties	Property	Value	

	Type	Variable: enumeration
	Label	Config Variant
	Default	VariantPostBuild
	Range	VariantPostBuild VariantPreCompile
Returns	N/A	

3.2.2 MbConfigSet

Container	MbConfigSet	
Description	This container contains the configuration parameters and sub containers of the Mb module.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.2.1 MbMaxChannel

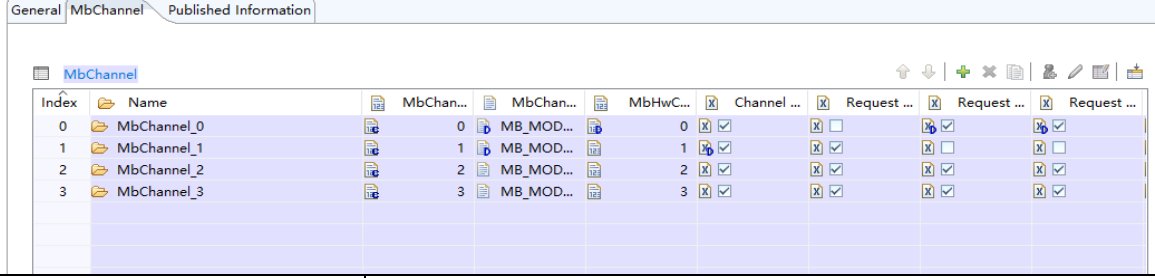
Variable	MbMaxChannel	
Description	The value for the MbMaxChannel must match with the number of MbChannel configured	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.2.2 MbGetSpinLockTimeout

Variable	MbGetSpinLockTimeout	
Description	The timeout for acquiring a spin lock.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.2.3 MbChannel

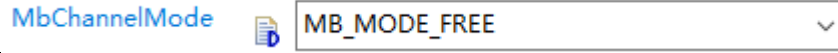
Container	MbChannel	
Description	Configuration of an individual Mailbox channel.	

Screenshot		
Properties	Property	Value
	Type	List
	MIN	1

3.2.2.3.1.1 MbChannelId

Container	MbChannelId	
Description	Channel Id of the Mailbox channel. This value will be assigned to the symbolic name derived of the MbChannel container short name.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	true
	Range	0 -> 14

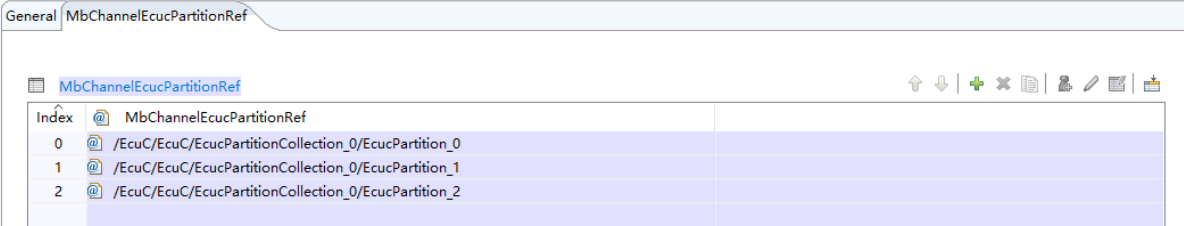
3.2.2.3.2 MbChannelMode

Variable	MbChannelMode	
Description	Configures the mode of this channel.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	MB_MODE_FREE, MB_MODE_MESSAGE_QUEUE

3.2.2.3.3 MbHwChannel

Variable	MbHwChannel	
Description	Hardware channel of the Mailbox channel.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	1 -> 4294967295

3.2.2.3.4 MbChannelEcucPartitionRef

Variable	MbChannelEcucPartitionRef	
Description	Maps the Mailbox Channel to zero or multiple ECUC partitions. The ECUC partitions referenced are a subset of the ECUC partitions where the Mb driver is mapped to.	
Screenshot		
Properties	Property	Value
	Type	Variable: REFERENCE
	Origin	FLAGCHIP
	Ref	ASPathDataOfSchema:/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

3.2.2.3.5 MbFreeChannel

Variable	MbFreeChannel	
Description	This container contains the configuration (parameters) in case the mode is "Free Channel"	
Screenshot		
Properties	Property	Value
	Type	Container
	Origin	FLAGCHIP

3.2.2.3.5.1 MbChannelAutoDone

Variable	MbChannelAutoDone	
Description	Enable / disable the auto done of the channel.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.2.3.5.2 MbChannelReqToCore0

Variable	MbChannelReqToCore0	
Description	Enable / disable the request to Core0 for this channel.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP

	SymbolicNameValue	false
--	-------------------	-------

3.2.2.3.5.3 MbChannelReqToCore1

Variable	MbChannelReqToCore1	
Description	Enable / disable the request to Core1 for this channel.	
Screenshot	Request To Core1  ☒ 	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.2.3.5.4 MbChannelReqToCore2

Variable	MbChannelReqToCore2	
Description	Enable / disable the request to Core2 for this channel.	
Screenshot	Request To Core2  ☒ 	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.2.3.5.5 MbChannelDoneToCore0

Variable	MbChannelDoneToCore0	
Description	Enable / disable the done to Core0 for this channel.	
Screenshot	Done To Core0  ☒ 	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.2.3.5.6 MbChannelDoneToCore1

Variable	MbChannelDoneToCore1	
Description	Enable / disable the done to Core1 for this channel.	
Screenshot	Done To Core1  ☐ 	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

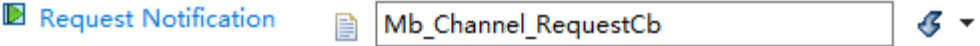
3.2.2.3.5.7 MbChannelDoneToCore2

Variable	MbChannelDoneToCore2	
Description	Enable / disable the done to Core2 for this channel.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

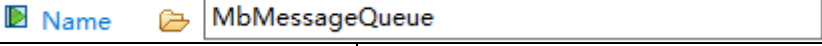
3.2.2.3.5.8 MbRequestNotification

Variable	MbRequestNotification	
Description	Notification function for free channel request event.	
Screenshot		
Properties	Property	Value
	Type	Variable: FUNCTION-NAME
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Optional	true

3.2.2.3.5.9 MbDoneNotification

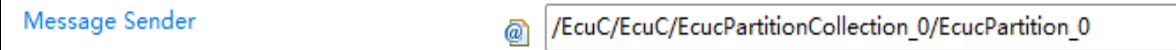
Variable	MbDoneNotification	
Description	Notification function for free channel done event.	
Screenshot		
Properties	Property	Value
	Type	Variable: FUNCTION-NAME
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Optional	true

3.2.2.3.6 MbMessageQueue

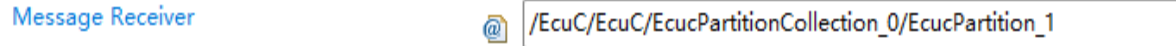
Variable	MbMessageQueue	
Description	This container contains the configuration (parameters) in case the mode is "Message Queue"	
Screenshot		
Properties	Property	Value
	Type	Container
	Origin	FLAGCHIP

3.2.2.3.6.1 MbMsgSender

Variable	MbMsgSender
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Description	Select the core responsible for sending messages.	
Screenshot		
Properties	Property	Value
	Type	Variable: REFERENCE
	Origin	FLAGCHIP
	Ref	ASPathDataOfSchema: /AUTOSAR/EcuDefs/EcuC/EcuPartitionCollection/EcuPartition

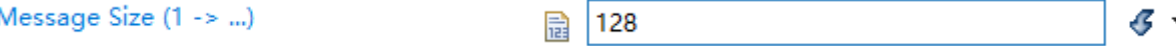
3.2.2.3.6.2 MbMsgReceiver

Variable	MbMsgReceiver	
Description	Select the core responsible for receiving messages.	
Screenshot		
Properties	Property	Value
	Type	Variable: REFERENCE
	Origin	FLAGCHIP
	Ref	ASPathDataOfSchema: /AUTOSAR/EcuDefs/EcuC/EcuPartitionCollection/EcuPartition

3.2.2.3.6.3 MbMsgBlockNum

Variable	MbMsgBlockNum	
Description	Message block number of the Mb channel.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	1->...

3.2.2.3.6.4 MbMsgBlockNum

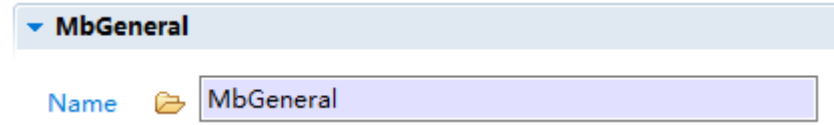
Variable	MbMsgBlockNum	
Description	Buffer size of each message.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	1->...

3.2.2.3.6.5 MbReceivedNotification

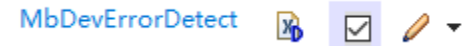
Variable	MbReceivedNotification	
-----------------	------------------------	--

Description	Notification function for receiving message.	
Screenshot		
Properties	Property	Value
	Type	Variable: FUNCTION-NAME
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Optional	true

3.2.3 MbGeneral

Container	MbGeneral	
Description	Configuration of general Mb parameters.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.1 MbDevErrorDetect

Variable	MbDevErrorDetect	
Description	Switches the Development Error Detection and Notification on or off.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.3.2 MbMulticoreSupport

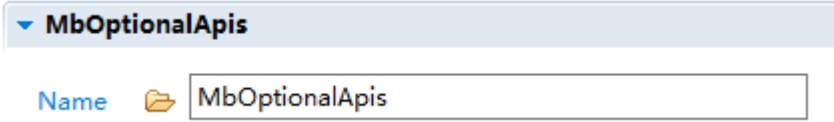
Variable	MbMulticoreSupport	
Description	Switches the multicore support of Mailbox module.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.3.1 MbDisableRuntimeErrorDetect

Variable	MbDisableRuntimeErrorDetect	
Description	Switches to Disable Runtime Error Detection and Notification on or off.	
Screenshot		
Properties	Property	Value

	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.4 MbOptionalApis

Container	MbOptionalApis	
Description	This container contains all configuration switches for configuring optional API services of the Mailbox driver.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.4.1 MbDelInitApi

Variable	MbDelInitApi	
Description	Adds / removes the service Mb_DelInit() from the code.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.4.2 MbFreeChannelApi

Variable	MbFreeChannelApi	
Description	Adds / removes the service Mb_DoneChannel(),Mb_SendData() from the code.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.4.3 MbMessageQueueApi

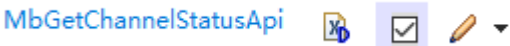
Variable	MbMessageQueueApi	
Description	Adds / removes the service Mb_SendMessage(),Mb_GetMessage(),Mb_GetMessageCount(),Mb_GetMessageQueueStatus(),Mb_ResetMessageQueue() from the code.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP

	SymbolicNameValue	false
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3.2.4.4 MbResetChannelApi

Variable	MbResetChannelApi	
Description	Adds / removes the service Mb_ResetChannel() from the code.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

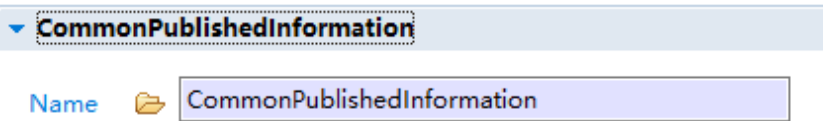
3.2.4.5 MbGetChannelStatusApi

Variable	MbGetChannelStatusApi	
Description	Adds / removes the service Mb_GetChannelStatus() from the code.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.4.6 MbGetVersionInfoApi

Variable	MbGetVersionInfoApi	
Description	Adds / removes the service Mb_GetVersionInfo() from the code.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

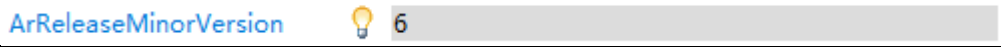
3.2.5 CommonPublishedInformation

Container	CommonPublishedInformation	
Description	Common container, aggregated by all modules. It contains published information about vendor and versions.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.5.1 ArReleaseMajorVersion

Variable	ArReleaseMajorVersion	
Description	Major version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	4

3.2.5.2 ArReleaseMinorVersion

Variable	ArReleaseMinorVersion	
Description	Minor version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	6

3.2.5.3 ArReleaseRevisionVersion

Variable	ArReleaseRevisionVersion	
Description	Revision version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0

3.2.5.4 SwMajorVersion

Variable	SwMajorVersion	
Description	Major version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0

3.2.5.5 SwMinorVersion

Variable	SwMinorVersion	
Description	Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	6

3.2.5.6 SwPatchVersion

Variable	SwPatchVersion	
Description	Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0

3.2.5.7 ModuleId

Variable	ModuleId	
Description	Module ID of this module from Module List.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	254

3.2.5.8 VendorId

Variable	VendorId	
Description	Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	174

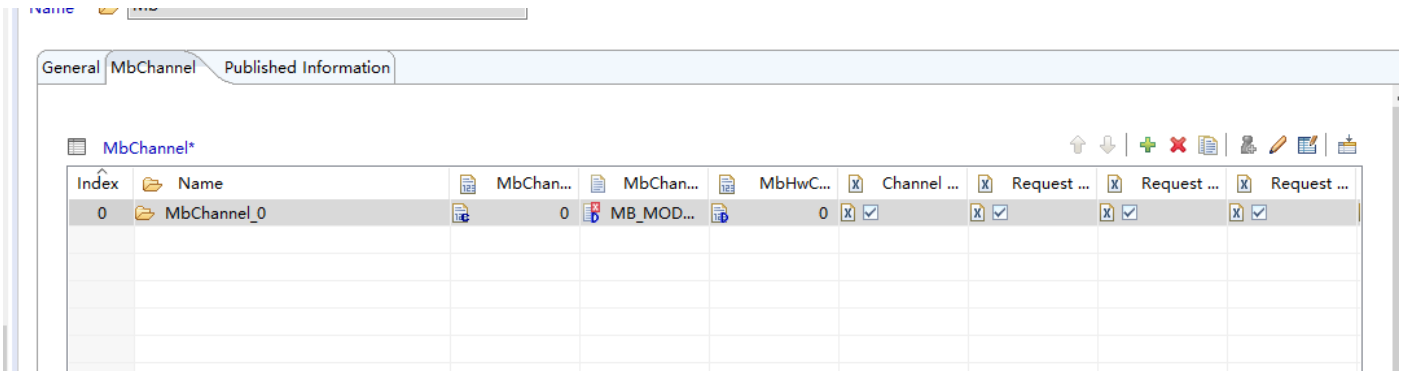
Chapter 4 Configuration Guides

4.1 Configuration Item Constraint

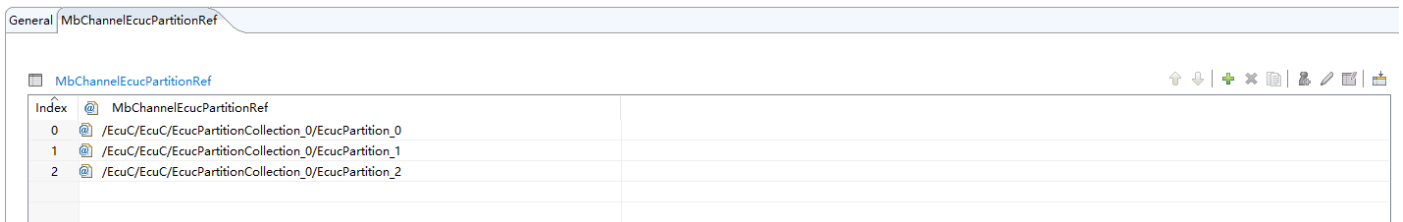
- 1) Each Mailbox channel must use a different physical channel.
- 2) When MbMulticoreSupport is not enabled, the Mailbox channel can only be set to Free Mode.

4.2 Mailbox Usage Common Steps

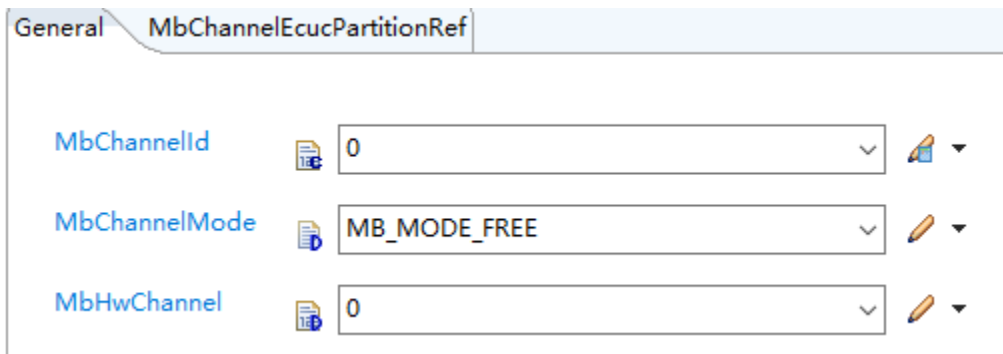
- 1) Configure a Mailbox channel.
 - a) Add an object in the MbChannel tab.



- b) Add partitions to the Mailbox channel. The partition referred from the EcuC module.



- c) Configure the mode and physical channel of the Mailbox channel.

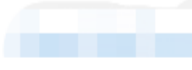


- d) When set to Free Mode.
 - i. Enable / disable the auto done of the channel.

- Enable: Channel will automatically done after receive a request.
- Disable: Channel needs to be manually done.

Channel Auto Done  ☒ 

- ii. Select which core is able to receive requests sent from this channel.

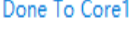
 Request To Core1  ☒  

 Request To Core0  ☐ 

 Request To Core2  ☒ 

- iii. Select which core is able to receive dones sent from this channel.

 Done To Core0  ☒ 

 Done To Core1  ☐ 

 Done To Core2  ☐ 

- iv. Set up notification functions for Request and Done

 Request Notification  Mb_Channel_RequestCb 

 Done Notification  Mb_Channel_DoneCb 

- e) When set to Message Queue.

- i. Select which core is allowed to send messages.

Message Sender  /EcuC/EcuC/EcucPartitionCollection_0/EcucPartition_0

- ii. Select which core is allowed to receive messages.

Message Receiver  /EcuC/EcuC/EcucPartitionCollection_0/EcucPartition_1

- iii. Configure the message block number of the message queue.

Message Buffer Number (1 -> ...)  10 

- iv. Configure the size of each blocks.

Message Size (1 -> ...)  128 

- v. Set up notification function for receiving message.

 Message Received Notification  Mb_Channel2_ReceivedCb 

2) General Configuration:

- a) Select the General tab.

- b) Input the number of objects to be added to the MbChannel tab.

MbMaxChannel  4 

- c) Input the timeout for acquiring a spin lock

[MbGetSpinLockTimeout \(1 -> 4294967295\)](#)

 65535 

- d) Enable or disable the development error detection.

[MbDevErrorDetect](#)  ☒ 

- e) Choose which API will be used.

MbOptionalApis

[Name](#) 

MbDeInitApi		☒		MbFreeChannelApi		☒	
MbMessageQueueApi		☒		MbResetChannelApi		☒	
MbGetChannelStatusApi		☒		MbGetVersionInfoApi		☒	

- 3) Generate configuration files.